

Advanced (Habanero)**Q1.** Which of the following is a relation but not a function?

- (A) $\{(1, 2), (2, 3), (3, 4)\}$
 (B) $\{(1, 2), (2, 3), (3, 2)\}$
 (C) $\{(1, 2), (1, 3), (2, 3)\}$
 (D) $\{(1, 1), (2, 2), (3, 3)\}$
 (E) $\{(1, 2), (2, 2), (3, 2)\}$

Q2. Which vertical line test result indicates a function?

- (A) More than one x -value
 (B) More than one y -value for each x
 (C) Exactly one y -value for each x
 (D) No y -values
 (E) More than one x -value for each y

Q3. What is the domain of the function $f(x) = \frac{1}{x}$?

- (A) $(-\infty, 0)$
 (B) $(-\infty, 0) \cup (0, \infty)$
 (C) $[0, \infty)$
 (D) $(-\infty, \infty)$
 (E) $[0, 1)$

Q4. The range of the function $g(x) = |x|$ is

- (A) $[0, \infty)$
 (B) $(-\infty, 0)$
 (C) $(-\infty, 0) \cup (0, \infty)$
 (D) $(-\infty, \infty)$
 (E) $[0, 1)$

Q5. What is $f(-3)$ if $f(x) = x^2 + 2$?

- (A) -9
 (B) -7
 (C) 7
 (D) 11
 (E) 15

Q6. If $h(x) = 2x - 1$, what is $h(2x)$?

- (A) $2x - 1$
 (B) $4x - 1$
 (C) $4x + 1$
 (D) $2x + 1$
 (E) $x - 2$

Q7. The equation $y = |x - 2|$ represents a

- (A) Linear function
 (B) Quadratic function
 (C) Absolute value function
 (D) Rational function
 (E) Exponential function

Q8. For the function $f(x) = 3x - 2$, what is $f(0)$?

- (A) -2
 (B) -1
 (C) 0
 (D) 1
 (E) 2

Open Ended Questions (FRQ)**Q9.** Determine the domain and range of the function $f(x) = \frac{x}{x^2 - 4}$ and explain your reasoning.**Q10.** Consider the function $g(x) = |2x - 1|$. Evaluate $g(2)$ and $g(-1)$, then graph the function. Label the x - and y -axes and provide a title for the graph.**Chef's Tips**

- Remind students to clearly label their graphs and axes.
- Emphasize the importance of showing all work for free response questions.
- Encourage students to use proper notation when evaluating functions.